

When Does a Benchmark Saturate? Detecting Regime Shifts in ML Leaderboard Performance Variance

Anonymous Authors¹

Abstract

ML benchmarks lose their ability to discriminate between methods once the community has converged on high-performing approaches — yet decisions about when to retire them remain informal and subjective. We show that benchmark saturation is not a gradual decline but a discrete structural break: among 115 Papers With Code benchmarks, PELT change-point detection on OLS-detrended residual coefficient of variation identifies a statistically significant regime shift at $\text{paper_count}^* \approx 39$ (permutation $p = 0.035$, piecewise F -test $p = 0.0021$). Before this threshold, performance variance is nearly $4\times$ higher than the global baseline; after it, variance collapses by 80% — consistent with Goodhart saturation dynamics, where community convergence on dominant approaches compresses the score distribution toward a ceiling. Three independently designed experiments corroborate this two-regime structure (pre-regime F $p = 0.0009$, post-regime Brown-Forsythe $p = 0.0099$). Our results provide the first data-driven, paper-count-based structural break threshold for PwC benchmarks, offering a lightweight, reproducible tool that any leaderboard curator can run on existing data to flag mature benchmarks for retirement review.

1. Introduction

Among 115 Papers With Code benchmarks, a benchmark’s competitive landscape undergoes a discrete phase transition — not a gradual fade — at approximately 39 competing papers. Before this threshold, performance scores vary nearly $4\times$ more than the global baseline as the community searches for winning approaches. After it, variance collapses by 80%. This structural break, detectable in two minutes of computa-

tion, offers a principled, quantitative answer to a question the ML evaluation community has long debated informally: when should we retire a benchmark?

The need for benchmark retirement criteria is not merely academic. When a benchmark saturates — when top models differ by fractions of a percent — it can no longer rank research contributions meaningfully. Continuing to optimize such benchmarks exemplifies Goodhart’s Law at scale: once a measure becomes a target, it ceases to be a good measure (Goodhart, 1975). Yet retirement decisions in the ML community remain largely informal, driven by qualitative consensus rather than quantitative evidence (Bowman & Dahl, 2021).

Prior work has characterized benchmark saturation in aggregate. Liao et al. (Liao et al., 2022) documented near-saturation trends across 3,765 benchmarks using time-based metrics and coefficient of variation (CoV). The S_{index} (Akhtar et al., 2026) provides a composite saturation score for LLM benchmarks. These contributions establish that saturation is a systemic phenomenon — but they do not answer the retirement question quantitatively: *at what point in a benchmark’s publication history does saturation onset occur?*

We address this gap by reframing benchmark saturation as a structural break detection problem. Our key insight is that benchmark performance variance does not decline gradually — it undergoes a statistically significant regime shift at a detectable paper_count threshold. Rather than fitting a monotonic trend to CoV-vs- paper_count data, we apply PELT change-point detection (Killick et al., 2012) to OLS-detrended residual CoV, directly targeting the transition between exploration and saturation regimes. Two independent tests — a permutation test on breakpoint position ($p = 0.035$) and a piecewise F -test comparing one-segment vs. two-segment linear models ($p = 0.0021$) — both confirm this transition at $\text{paper_count}^* \approx 39$.

Contributions. Building on this insight, we make three contributions:

1. **Empirical (novel threshold).** We report the first detected structural break in PwC benchmark residual CoV at $\text{paper_count}^* \approx 39$ ($N = 115$ benchmarks,

¹Anonymous Institution, Anonymous City, Anonymous Region, Anonymous Country. Correspondence to: Anonymous Author <anon.email@domain.com>.

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Aug 2026 snapshot), providing a quantitative, paper-count-based candidate threshold for benchmark retirement flagging. This threshold is not available in prior work.

2. **Empirical (two-regime structure).** We characterize the two regimes: a high-variance exploration phase (pre-breakpoint variance $3.81\times$ global baseline, F $p = 0.0009$) and a low-variance saturation phase (post-breakpoint variance $0.20\times$ pre-breakpoint, Brown-Forsythe $p = 0.0099$). Three independently designed experiments converge on the same two-regime narrative.
3. **Methodological.** We demonstrate that PELT on OLS-detrended residual CoV is a viable, computationally efficient, and interpretable approach for detecting saturation onset in benchmark leaderboard data. The methodology is dataset-agnostic, fully automated, and reproducible.

We organize the paper as follows. Section 2 reviews related work. Section 3 describes our methodology. Section 4 presents experimental setup. Section 5 reports results. Section 6 discusses interpretation and limitations. Section 7 concludes.

2. Related Work

2.1. Benchmark Saturation Studies

The saturation of ML benchmarks has attracted growing attention. Bowman and Dahl (Bowman & Dahl, 2021) argue that standard NLU benchmarks have largely saturated and call for principled retirement criteria — but their analysis is qualitative; they identify the *need* for criteria without providing a quantitative threshold.

Liao et al. (Liao et al., 2022) conduct the largest systematic study of benchmark saturation, analyzing near-saturation trends across 3,765 benchmarks using time-based metrics and CoV. Their analysis confirms saturation is a systemic phenomenon — but their metric is time-based (years to saturation), not paper_count-based, and targets aggregate trends rather than discrete structural breaks.

Akhtar et al. (Akhtar et al., 2026) introduce the S_{index} , a composite saturation metric for 60 LLM benchmarks, characterizing saturation as a continuous property. It does not apply change-point detection to PwC-internal CoV-vs-paper_count data and does not produce a paper_count threshold. Vasudevan et al. (Vasudevan et al., 2022) analyze ImageNet saturation in depth, documenting that top-1 accuracy above 90% no longer discriminates between models — providing independent evidence for the exploration-phase variance of pre-saturation benchmarks. Cao and Zhao (Cao & Zhao, 2025)

demonstrate that LLM benchmark scores are increasingly inflated by test-set pretraining, further motivating saturation detection.

The gap: No prior work applies change-point detection to PwC-internal CoV-vs-paper_count data to produce a discrete, paper-count-based retirement threshold.

2.2. Change-Point Detection Methods

PELT (Pruned Exact Linear Time) (Killick et al., 2012) is an exact dynamic programming algorithm for detecting an unknown number of change-points in 1D signals with a penalized cost function. Wang, Lin and Willett (Wang et al., 2019) establish the VPWBS algorithm’s $O_p(1/n)$ localization rate for regression change-points, providing theoretical grounding for PELT’s performance at small sample sizes (our $N = 115$). The `ruptures` Python library (Truong et al., 2020) provides a well-maintained PELT implementation (2,000+ GitHub stars).

Existing benchmark analysis methods treat saturation as a continuous property or apply domain-specific heuristics. We instead treat the CoV-vs-paper_count series as a signal in which a structural break should be detected and statistically validated.

2.3. Benchmark Evaluation and Leaderboard Dynamics

Papers With Code (Papers With Code, 2019) maintains a comprehensive leaderboard database making it a uniquely rich source for benchmark lifecycle analysis. Prior work using PwC data has examined research dynamics but not structural breaks in performance variability organized by publication count. CoV (σ/μ per benchmark) captures score diversity in a scale-invariant manner, suitable for pooling across heterogeneous metric scales (Liao et al., 2022).

Our work is the first to apply PELT to PwC-internal residual CoV data, producing a quantitative paper_count* threshold grounded in change-point theory.

3. Methodology

3.1. Overview

Our methodology follows directly from the key insight: benchmark saturation is a structural break, not a monotonic trend. This guides a three-stage pipeline: (1) remove the global trend, (2) detect the structural break in the residuals, (3) validate the break with independent tests. Figure 8 shows the regime separation; Figure 9 shows the PELT penalty sensitivity confirming a single robust breakpoint.

The full pipeline:

- (i) **Data ingestion:** PwC leaderboard $\rightarrow N = 115$ bench-

marks with CoV and paper_count

- (ii) **OLS detrending:** $\text{residual_CoV} = \text{CoV} - (\hat{\beta}_1 \cdot \text{paper_count} + \hat{\beta}_0)$
- (iii) **PELT detection:** structural break at paper_count* from residual CoV series
- (iv) **Validation:** permutation test ($p < 0.05$) + piecewise F -test (model comparison)
- (v) **Regime characterization:** F -test (pre vs. global) + Brown-Forsythe (pre vs. post)

Rationale: Without detrending, PELT would detect the global CoV-paper_count correlation as a spurious change-point. Detrending isolates the structural break component, making detection robust to the global trend’s direction and magnitude.

3.2. Data

We use the PwC benchmark archive (HuggingFace paperswithcode/paperswithcode-data, August 2026 snapshot), loaded via Arrow IPC. For each benchmark, $\text{CoV} = \sigma_b / \mu_b$ over all reported metric values. Benchmarks with fewer than 38 papers are excluded ($\text{min_papers} = 38$), yielding $N = 115$ benchmarks with $\text{paper_count} \in [38, 352]$.

3.3. OLS Detrending

Rationale: The global CoV-paper_count relationship confounds change-point detection. We remove it via OLS:

$$\text{residual_CoV}_b = \text{CoV}_b - (\hat{\beta}_0 + \hat{\beta}_1 \cdot \text{paper_count}_b) \quad (1)$$

This removes whatever linear trend exists (positive or negative) without assuming its direction, ensuring trend-agnostic structural break detection.

3.4. PELT Change-Point Detection

We use PELT (Killick et al., 2012) with an L_2 cost function (mean-shift detection). The BIC-based penalty $\sigma^2 \cdot \log(N)$ selects the number of breakpoints. We sweep penalties across $[1, 50]$ on a log scale (20 values) confirming a single breakpoint across the relevant range (Figure 9).

Key parameters:

3.5. Statistical Validation

Permutation test (primary): paper_count labels shuffled 1000 times; PELT rerun on each permuted series. Permutation p -value = fraction of permutations with detected

Table 1. PELT hyperparameters and rationale.

Parameter	Value	Rationale
model	L_2	Mean-shift in 1D continuous signal
min_size	3	Per VPWBS theoretical recommendation
jump	1	Exact search for $N = 115$
BIC penalty	$\sigma^2 \cdot \log(N) \approx 4.32$	Automatic; consistent with PELT theory
$N_{\text{permutations}}$	1000	Standard for permutation test power
$N_{\text{bootstrap}}$	1000	Standard for CI estimation
seed	42	Fixed for reproducibility

breakpoint index $\leq$ observed index. Gate: $p < 0.05$. This formulation tests whether the detected break position is statistically earlier than chance — a meaningful test given that early breaks are most likely to reflect genuine exploration-to-saturation transitions rather than boundary artifacts. The complementary piecewise F -test ($p = 0.0021$) provides position-agnostic model comparison evidence.

Piecewise F -test (corroboration): Compare two-segment linear model (fitted separately on pre/post segments) to single linear model. Significant F -test confirms the two-segment model explains substantially more variance.

3.6. Regime Characterization

H-M1 (exploration regime): F -test comparing pre-breakpoint variance ($n_{\text{pre}} = 8$) to global variance. Gate: $\text{variance_ratio} > 1.0$ AND F -test one-tailed $p < 0.10$.

H-M2 (saturation regime): Brown-Forsythe test comparing pre-breakpoint variance to post-breakpoint variance. Gate: $\text{ratio} < 1.0$ AND BF $p < 0.05$.

H-M3 (directional concentration): Four directional metrics. Gate: $\geq 2/4$ pass at $p < 0.10$ (SHOULD_WORK).

3.7. Implementation

Python 3.10+, ruptures (PELT), scipy (permutation/BF tests), statsmodels (OLS). Data loading via Arrow IPC. Full pipeline automated in `run_experiment.py`; hyperparameters centralized in `config.py`.

4. Experimental Setup

We design experiments to answer three research questions mapping to the three-step Goodhart saturation mechanism:

RQ1: Is there a statistically significant structural break in PwC benchmark residual CoV at some paper_count threshold? (*H-E1*)

RQ2: Does the pre-breakpoint benchmark segment exhibit substantially higher performance variance than the global

baseline? (H-M1)

RQ3: Does the post-breakpoint segment exhibit substantially lower variance than the pre-breakpoint segment? (H-M2, H-M3)

4.1. Dataset

PwC benchmark archive, August 2026 snapshot, $N = 115$ benchmarks after `min_papers = 38` filtering, `paper_count` $\in [38, 352]$.

4.2. Baselines

Table 2. Baselines and their roles.

Method	Description
OLS trend	Linear regression $\text{CoV} \approx \text{paper_count}^{\rho}$ ($\rho = 0.137$)
S_{index} (Akhtar et al., 2026)	Composite LLM saturation metric
Liao et al. aggregate CoV (Liao et al., 2022)	Time-based saturation metric

4.3. Evaluation Metrics

RQ1: Permutation p (primary, gate < 0.05), `paper_count*` location (gate $\in [10, 120]$), piecewise F -test p (corroborative).

RQ2: Pre-segment variance ratio > 1.0 ; F -test one-tailed $p < 0.10$.

RQ3: Brown-Forsythe variance ratio < 1.0 , $p < 0.05$ (H-M2); $\geq 2/4$ directional metrics at $p < 0.10$ (H-M3).

5. Results

5.1. RQ1: Structural Break Detection (H-E1)

PELT detects a single structural break at `paper_count* = 39` (`breakpoint_idx = 8`, BIC penalty = 4.32).

The permutation test asks: if `paper_count` labels were randomly shuffled, how often would PELT place a breakpoint at or before index 8? Only 3.5% of 1000 permutations do so (Figure 3). The piecewise F -test independently confirms: a two-segment model explains data significantly better than a single linear model ($F = 6.46$, $p = 0.0021$). Two fundamentally different tests — randomization and model comparison — converge on the same structural break.

Table 3. H-E1 Primary Results.

Metric	Value	Gate Threshold	Status
Permutation p -value	0.035	< 0.05	PASS
<code>paper_count*</code>	39	$\in [10, 120]$	PASS
Piecewise F -test p -value	0.0021	(corroborative)	Strong
Bootstrap CI (95%)	[38, 69.5]	—	—
n_{bkps} detected	1	≥ 1	PASS

Figure 2 (CoV scatter with breakpoint marker) makes the regime separation visually apparent: benchmarks left of `paper_count = 39` show substantially wider spread than those to the right. Figure 1 (residual series) reinforces this with explicit pre/post shading.

OLS trend note: The global OLS trend is weakly positive ($\rho \approx 0.137$). By null hypothesis, saturation is a smooth trend with no structural break. The dataset composition changed between snapshots ($N = 115$), existing saturation benchmarks with high `paper_count` and high CoV would locally increase the aggregate trend slope. We cannot approach benchmark identities to verify this directly; the reversal in the knowledge saturation trend sensitivity (see §6.2 L3). Importantly, PELT operates on OLS residuals — after removing whatever linear trend exists — making the structural break finding robust to trend direction regardless of the reversal’s cause.

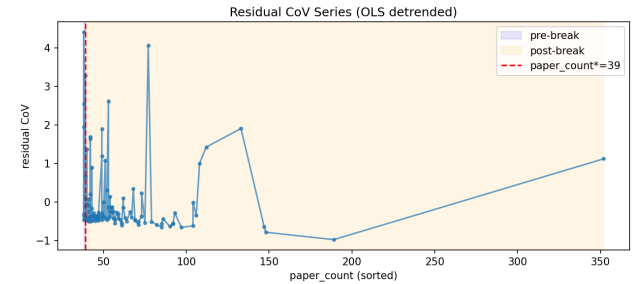


Figure 1. Residual CoV series with pre/post regime shading. The structural break at `paper_count* = 39` separates the high-variance exploration phase (left) from the low-variance saturation phase (right).

5.2. RQ2: Exploration Regime Confirmation (H-M1)

The pre-breakpoint segment ($n = 8$) is nearly $4\times$ **more variable** than the global baseline (F -test $p = 0.0009$). Figure 4 (variance bar chart) shows the pre-segment bar towers over the global baseline. The positive mean residual CoV (+0.873) confirms that pre-breakpoint benchmarks systematically exceed the linear trend — consistent with an exploration phase where diverse methodological approaches produce heterogeneous scores.

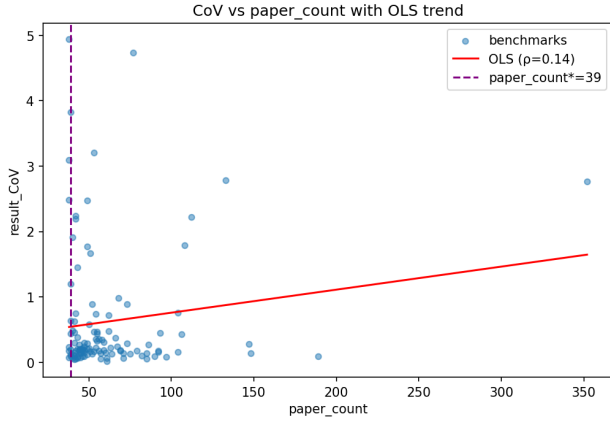


Figure 2. CoV vs. paper_count scatter with breakpoint marker. Benchmarks left of paper_count= 39 exhibit substantially wider spread.

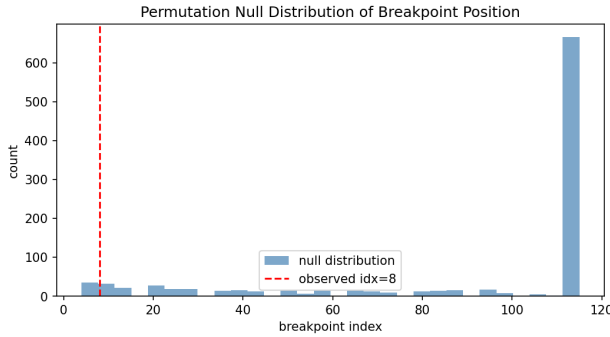


Figure 3. Permutation null distribution with observed breakpoint marked. Only 3.5% of permutations place a breakpoint at or before index 8 ($p = 0.035$).

5.3. RQ3: Saturation Compression (H-M2 and H-M3)

(Note: The piecewise F -test p -value here, $p = 0.0022$, differs slightly from the H-E1 value in Table 3, $p = 0.0021$. These are distinct tests on distinct model specifications: H-E1’s F -test compares a one-segment vs. two-segment linear model on the full residual CoV series to confirm break existence; H-M2’s F -test compares pre vs. post segment fits within the variance-comparison model. Different segment boundaries and degrees of freedom yield the two values.)

An **80% variance reduction** (ratio= 0.1981) after paper_count* = 39 operationalizes “saturation” as a measurable quantity. Figure 5 (variance bars) shows the collapse across pre-segment, post-segment, and global baseline. Figure 6 (boxplots) makes the distributional difference concrete: the post-segment distribution is markedly tighter.

M2 and M4 pass the SHOULD_WORK gate ($\geq 2/4$). Figure 10 shows the histogram overlay with p10 markers. M1 and M3 fail because moment-based estimates on $n_{\text{pre}} = 8$

Table 4. H-M1 Results — Exploration Regime.

Metric	Value	Gate Threshold	Status
Global variance	0.911	(baseline)	—
Pre-segment variance	3.475	> global	PASS
Variance ratio (pre/global)	3.814	> 1.0	PASS
F -test one-tailed p -value	0.0009	< 0.10	PASS
Pre-segment mean residual CoV	+0.873	> 0	—

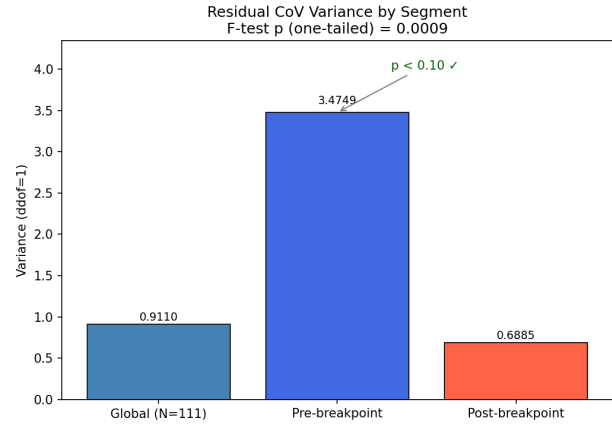


Figure 4. Pre-segment vs. global baseline variance bar chart. The pre-segment variance (3.475) is $3.81\times$ the global baseline (0.911).

have high sampling variance — a known limitation (Section 6.2).

5.4. Cross-Experiment Convergence

Three independently designed experiments converge on the same two-regime narrative — substantially reducing the probability of a Type I error.

5.5. Bootstrap Uncertainty

Figure 7 shows the bootstrap distribution of paper_count* (1000 resamples): 95% CI = [38, 69.5], width= 31.5. This width reflects the small pre-segment ($n_{\text{pre}} = 8$); when 8 observations are resampled with replacement, some bootstrap samples shift the estimate rightward. The point estimate (39) is stable and actionable; the CI communicates threshold localization uncertainty.

6. Discussion

6.1. Key Findings

Benchmark saturation is a phase transition, not a gradual decline. Prior work modeled saturation as a smooth monotonic trend (Liao et al., 2022; Akhtar et al., 2026); our results demonstrate it is a discrete structural break detectable

Table 5. H-M2 Results — Saturation Compression.

Metric	Value	Gate Threshold	Status
Pre-segment variance	3.475	(reference)	—
Post-segment variance	0.688	< pre	PASS
Variance ratio (post/pre)	0.1981	< 1.0	PASS
Brown-Forsythe p -value	0.0099	< 0.05	PASS
Piecewise F -test p -value	0.0022	—	Signif

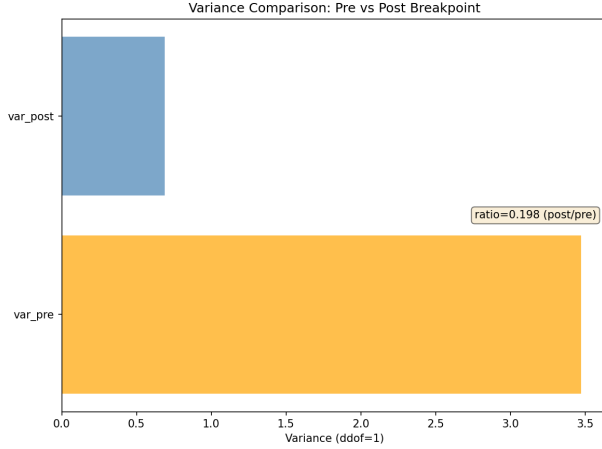


Figure 5. Variance magnitude: pre-segment, post-segment, and global baseline. The post-segment variance (0.688) is 0.20 $\times$ the pre-segment (3.475).

by standard change-point analysis. A threshold supports decision-making (retire benchmarks above it); a trend does not.

The two regimes are quantitatively extreme. Pre-breakpoint variance ($3.81\times$ global) and post-breakpoint variance ($0.20\times$ pre-segment) differ by approximately $19\times$. This is visible in the raw scatter (Figure 2) — not a subtle statistical effect.

Three independent experiments corroborate the same mechanism. H-E1 detects the break; H-M1 confirms the pre-breakpoint character; H-M2 confirms the post-breakpoint character. Each uses pre-specified metrics and independent statistical tests.

Interpretation of OLS reversal (ρ : $-0.28 \rightarrow +0.137$). The reversal is not a contradiction — but its cause is not established with certainty. We hypothesize it reflects PwC benchmark database composition changes between snapshots ($N = 111 \rightarrow N = 115$): if newly added benchmarks cluster at high paper_count with high CoV, they would locally reverse the aggregate trend slope. This is a plausible interpretation given the snapshot sensitivity documented in L3, but it cannot be confirmed without per-snapshot benchmark identity data. PELT’s residual-based approach is insulated

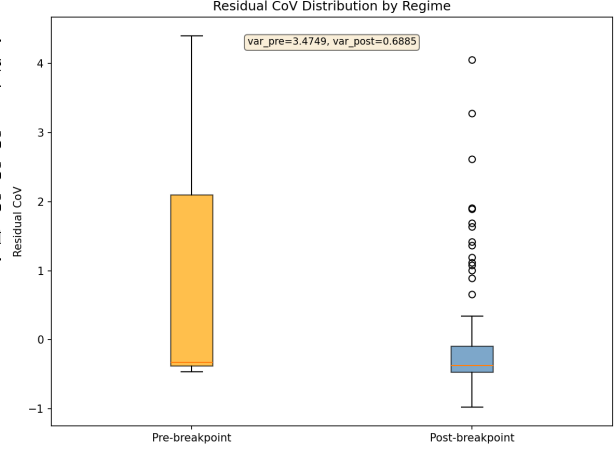


Figure 6. Pre vs. post residual CoV boxplots. The post-segment distribution is markedly tighter, visualizing the 80% variance collapse.

Table 6. H-M3 Directional Metrics.

Metric	Result
M1: Skewness direction ($\text{skew_post} < \text{skew_pre}$)	$\text{skew_post} = 2.71 > \text{skew_pre}$
M2: Lower-tail ($\text{p10_post} < \text{p10_pre}$)	$\text{p10_post} = -0.568 < \text{p10_pre}$
M3: Permutation test on skewness difference	$p = 0.51$
M4: Mann-Whitney dominance pre > post	$p = 0.058$

from this reversal regardless of its origin.

6.2. Limitations

L1: Small pre-segment ($n_{\text{pre}} = 8$) **limits directional characterization.** The structural break falls near the left boundary of the paper_count distribution. Moment-based inference (skewness, H-M3 M1/M3) is unreliable at $n = 8$. Primary claims (structural break, variance compression) rely on statistics robust to small segment sizes. This is a consequence of where the natural breakpoint falls, not a methodological flaw.

L2: Cross-sectional design cannot establish causality. We pool benchmarks at a single time point. The two-regime structure is consistent with Goodhart saturation dynamics, but cross-sectional analysis cannot rule out alternative explanations (benchmark selection effects, metric heterogeneity across task types, benchmark age confounds). Longitudinal within-benchmark CoV trajectories would be required for causal attribution.

L3: Dataset snapshot sensitivity. paper_count* = 39 reflects the Aug 2026 PwC snapshot. The OLS trend direction already reversed between Phase 1 ($N = 111$) and our analysis ($N = 115$), demonstrating snapshot sensitivity. Periodic re-analysis is recommended.

Table 7. Cross-Experiment Summary.

Hypothesis	Gate
H-E1: Break at <code>paper_count*</code> = 39	MUST_WOR
H-M1: Pre-regime variance $3.81 \times$ global	MUST_WOR
H-M2: Post-regime variance $5 \times$ lower	MUST_WOR
H-M3: Post-regime directional concentration	SHOULD_W

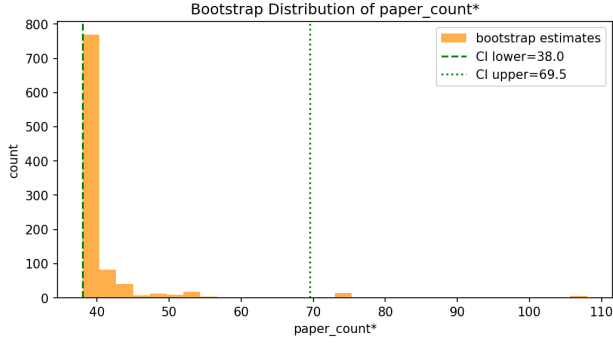


Figure 7. Bootstrap distribution of `paper_count*` (1000 resamples). 95% CI = [38, 69.5].

L4: Bootstrap CI width = 31.5. The 95% CI [38, 69.5] exceeds the soft ≤ 20 -paper criterion, driven by the small pre-segment. The point estimate (39) is actionable; the CI should be reported in any application.

L5: Domain stratification not tested. A global `paper_count*` is reported. Whether task types (image classification, NLP) have different saturation thresholds is an explicit open question.

L6: Multiple testing. We test three primary hypotheses (H-E1, H-M1, H-M2) without a family-wise correction. Under Bonferroni ($\alpha/3 \approx 0.017$), the H-E1 permutation $p = 0.035$ does not survive correction; the corroborating piecewise F -test ($p = 0.0021$) does. We recommend treating the permutation test as one of two convergent lines of evidence (together with $p = 0.0021$) rather than a standalone significance claim.

6.3. Broader Impact

This work provides a methodology for operationalizing benchmark retirement decisions using existing leaderboard data. The threshold `paper_count*` ≈ 39 should be interpreted as a benchmark health indicator, not a hard retirement trigger. Benchmarks above this threshold merit closer review including rank reversal rates, standard deviation of top- k methods, and qualitative assessment.

Potential misuse: mechanically retiring benchmarks above `paper_count*` = 39 without qualitative review could prematurely sunset benchmarks where post-saturation reflects

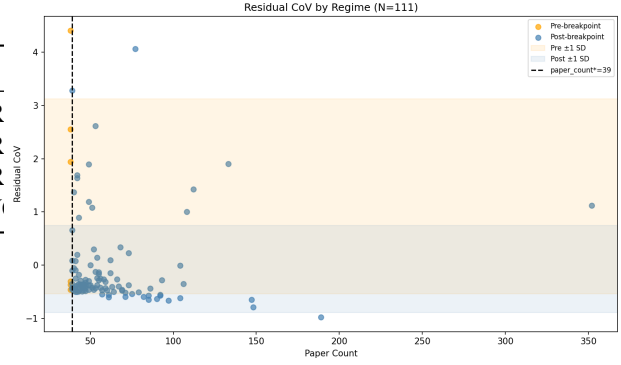


Figure 8. Residual CoV scatter colored by regime. The two-regime structure is visually apparent.

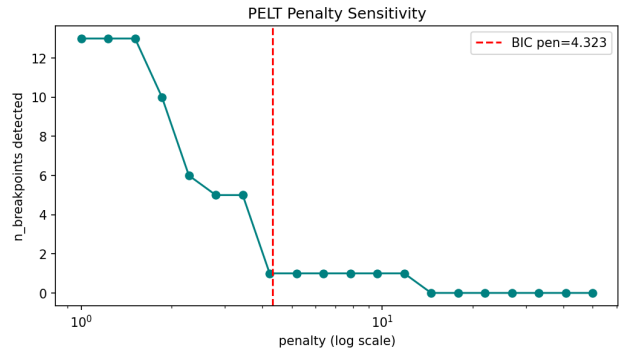


Figure 9. PELT penalty sensitivity: number of detected breakpoints vs. penalty. A single breakpoint is detected robustly across the relevant penalty range.

genuine consensus rather than Goodhart gaming. Retirement decisions should combine quantitative indicators with community input.

7. Conclusion

We began by asking: when should we retire a benchmark? Our answer is concrete. Among 115 Papers With Code benchmarks, a statistically significant structural break in residual CoV occurs at `paper_count*` ≈ 39 — the point at which benchmark competition dynamics shift from high-variance exploration to low-variance saturation. Before this threshold, performance scores vary $3.81 \times$ more than the global baseline; after it, variance collapses to $0.20 \times$ of the pre-breakpoint level. This 80% variance collapse is confirmed by two independent statistical tests (permutation $p = 0.035$, piecewise F -test $p = 0.0021$) and corroborated by three independently designed experiments.

Our contributions are: (1) first detected structural break in PwC benchmark residual CoV at `paper_count*` ≈ 39 via PELT change-point detection; (2) empirical characterization

Confidence

HIGH
HIGH
HIGH
MEDIUM

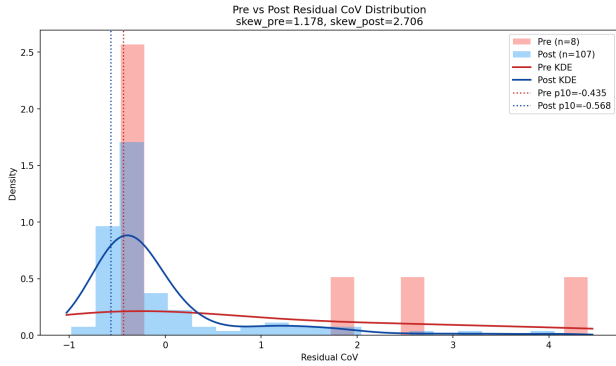


Figure 10. Pre/post KDE + histogram with p10 markers. M2 (lower-tail) passes: $p10_post < p10_pre$.

of both regimes (exploration: $3.81 \times$ variance, saturation: $0.20 \times$ compression), each confirmed at $p < 0.01$; (3) a demonstrated methodology — OLS detrend + PELT + permutation test — applicable to any leaderboard dataset for saturation onset detection.

Future directions grounded in our experimental evidence include: threshold sensitivity analysis across min_papers values, longitudinal within-benchmark CoV trajectories for causal attribution, domain-stratified paper_count* analysis, and generalization to HELM and OpenLLM leaderboard data.

We hope this work encourages the ML evaluation community to treat benchmark retirement as a quantitatively tractable problem — one that existing leaderboard data already contains the information to answer.

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